

Barcode Based Student Attendance System

A course project report submitted in partial fulfilment of the requirement
of

SMART SYSTEM DESIGN

by

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ABSTRACT

It is a “Barcode based Student Attendance System”, which works according to the MH-ET Live scanner as a command. Student attendance play significant role in order to justify academic outcome of a student and college as overall. Unfortunately, there is no automated attendance recordkeeping application available in colleges. There is a need for a tool to systematically keep the students attendance record due to increasing number of college students. The project that we are going to make is to help the teachers in our college to avoid maintaining the registry book. This project uses a barcode scanner. B.B.S.A.S uses Barcode scanner to take the attendance of students entering the lab. Each student’s ID card will have a barcode at the back side of it. This barcode contains unique data of the student such as roll number, branch and year. Etc. Student will scan their barcode at the end so that the student can’t cheat. The display screen will show the attendance of the particular student after scanning his/her barcode. Teachers and administrator will only have access to the system with their respective login ID’s and passwords.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Our aim is to make a Barcode based attendance system. Above the idea is to create attendance system with barcode where the information will be saved in excel sheet. These days attendance is mandatory for every student to attempt an exam. So this barcode-based attendance system. Usually, this can help the teachers out and save the time. Nowadays, technology is really influential and it is the core elements in almost educational institutions. They started to initiate the organisation with technology. Usually, the institutions used manual process for recording attendance by using paper and pen. It was one approach that has been used in almost institutions even the process seems slow and unproductive. Most of the attendance system is used manually which every student has to sign the attendance sheet paper and the lecturer needs to print new attendance sheet every week.

The project is a system that take down the student's attendance using Barcode. Every student is provided with a card containing unique barcode. Each barcode represents a unique id of students. Students should scan their id cards using barcode scanner and the system notes down their attendance as per dates. System then stores the student's attendance records and generates defaulter list. It also generates an overall report in excel sheet for admin.

This system helps facilitate attendance information of a particular student in a particular class and it is under the supervision of lecturers to generate the barcode and to collect data of attendance.

1.2 OBJECTIVES

- The main objective of this project is
- To design an efficient computerised system to be used for lecturers, students and parents.
- To implement the system to facilitate the lectures managing student's attendance.
- To evaluate the efficiency of the system in order to notify parents regarding the student's attendance.

- Better security.
- Generate results quickly.
- Provides accurate and efficient data.

CHAPTER 2

PROJECT DESCRIPTION

2.1 BLOCK DIAGRAM OF THE PROJECT

As shown in the above schematic diagram it mainly consists of an Arduino, MH-ET LIVE scanner. The scanner is connected to the Arduino with two jumper wires. As we scan the ID card using barcode scanner then the Arduino sends the information to the system. And the information is saved in the excel sheet. The block diagram of the project is shown in fig. 2.1.

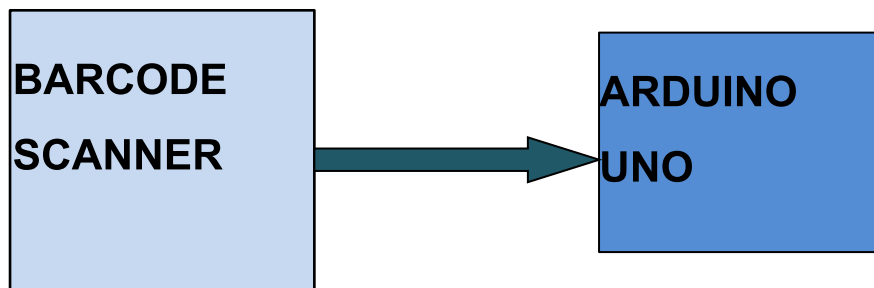


Fig.2.1 Block Diagram

2.2 HARDWARE DESCRIPTION

Arduino UNO

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 Analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode. Revision 3 of the board has the following new features: 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin, that is reserved for future purposes. Stronger RESET circuit. At mega 16U2 replace the 8U2. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the

reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see index of Arduino boards.

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)

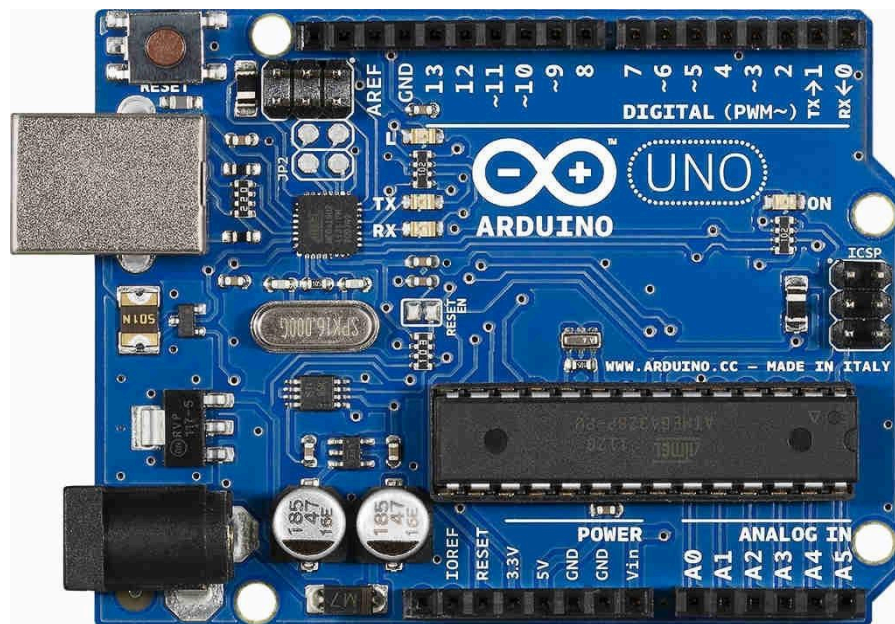


Fig. 2.2 Arduino Uno

Applications:

- Xoscillo, an open-source oscilloscope
- Arduinome, a MIDI controller device that mimics the Monome
- OBDuino, a trip computer that uses the on-board diagnostics interface found in most modern cars
- Gameduino, an Arduino shield to create retro 2D video games
- Arduino Phone, a do-it-yourself cell phone

- Water quality testing platform
- Automatic titration system based on Arduino and stepper motor
- Low cost data glove for virtual reality applications
- Impedance sensor system to detect bovine milk adulteration
- Homemade CNC using Arduino and DC motors with close loop control by Homofaciens
- DC motor control using Arduino and H-Bridge

BARCODE SCANNER

Scanner, also called optical scanner, computer input device that uses a light beam to scan codes, text, or graphic images directly into a computer or computer system. Barcode scanners are used widely at point-of-sale terminals in retail stores. A handheld scanner or bar-code pen is moved across the code, or the code itself is moved by hand across a scanner built into a checkout counter or other surface, and the computer stores or immediately processes

the data in the bar code. After identifying the product through its bar code, the computer determines its price and feeds that information into the cash register. Optical scanners are also used in fax machines and to input graphic material directly into personal computers. Flatbed scanners have a top that lifts up, and the into personal computer the material being scanned. Sheetfed scanners can be loaded with multiple loose pages, which are passed over the scanner by rollers. Barcode scanners record and translate barcodes from the image you recognize into alphanumeric digits.

The scanner then sends that information along to a computer database, either through a wired connection or wirelessly (depending on the model). Those digits refer to a particular item, and scanning the numbers and bars pulls up an entry in the database with further information such as the price, how many of this item in stock, a description of the item and possibly a picture for reference. Barcode scanners traditionally read the barcodes most people are familiar with. These include the 1D, or linear barcode, which most people are familiar with. These include the 1D, or linear barcode, which appears as an image of parallel lines and spaces (usually a quiet zone). A quiet zone is a space to prevent the reader from picking up other information that is not wanted or needed in the barcode scan. The blank space doesn't send any scanning signal and so it is referred to as a quiet space (or quiet zone). For many years, 1D barcodes were the standard, and they are still in widespread use today despite their smaller capacity for information (they can only contain

about 20-25 characters, though stacking the characters helps to increase that number). 2D barcodes store information both horizontally and vertically, resulting in exponentially larger storage capacity; however, an image scanner is required to read 2D barcodes, while a simple barcode scanner can only handle linear codes. There are different kinds of barcode scanners, some scanners use lasers, while others use lights or cameras, to capture the barcode image and turn it into an electronic code. There are pen readers, which have no moving parts, just a light source and a photodiode to measure the lines and spaces on the 1D barcode. Laser scanners use lasers as the light source, and have fewer errors scanning from a greater distance than pen readers (thanks to mirrors and lenses). CCD readers measure ambient light rather than self-reflected light, and take several readings per scan to reduce errors. Camera-based readers take a photo of the barcode in order to read and decode it, and they are a decent, cheaper option. The most advanced scanner of all is the omni-directional scanner, which is a laser scanner with more mirrors and lenses to reduce errors. An omni-directional scanner can read ripped, crumpled or otherwise damaged barcodes at a faster rate than other, lesser laser scanners.



Fig.2.3 MQ2 sensor

Features:

1. It is an input device that can read existing images directly. It converts the image in a data is in computer storable format and sends it to the computer for storage and processing.
2. A scanner also performs the job of Optical Character Recognition (OCR) using OCR software.

3. A scanner is useful with the fax cards in the system and it can scan and get the document.
4. Scanners are available in many shapes and size and their cost varies according to it.
5. A scanner needs 100KB to 8MB hard disk space to scanned images of simple black and white to a single high-quality, high-resolution colour image.
6. A scanner needs a VGA colour display with the system to view the scanned image.
7. A colour scanner supports up to 16.7million colours and can be programmed for simple black and white or grey shades images.
8. A scanner scans any image as a series of separate dots and it is capable to recognize minimum 300dots per inch i.e. The resolution starts from 300DPL.
9. The cheapest scanners are hand held scanner that are used to scan small figures into the computer.
10. The most expensive scanner are flatbed scanners and are capable of scanning any size image of any colour with the same efficiency and accuracy.
11. The flatbed scanner is connected to the computer through SCSI or the proprietary interface provided by the manufacture.

2.3 SOFTWARE DESCRIPTION

The software used here is ARDUINO SOFTWARE:

The Arduino Integrated Development Environment - or Arduino Software (IDE) contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

Writing Sketches:

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

NB: Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension pde. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the into extension on save.



Verify

Checks your code for errors compiling it.



Upload

Compiles your code and uploads it to the configured board. See uploading below for details.

Note: If you are using an external programmer with your board, you can hold down the "shift" key on your computer when using this icon. The text will change to "Upload using Programmer"



New

Creates a new sketch.



Open

Presents a menu of all the sketches in your sketchbook. Clicking one will open it within the current window overwriting its content.

Note: due to a bug in Java, this menu doesn't scroll; if you need to open a sketch late in the list, use the File | Sketchbook menu instead.



Save

Saves your sketch.



Serial Monitor

Opens the serial monitor.

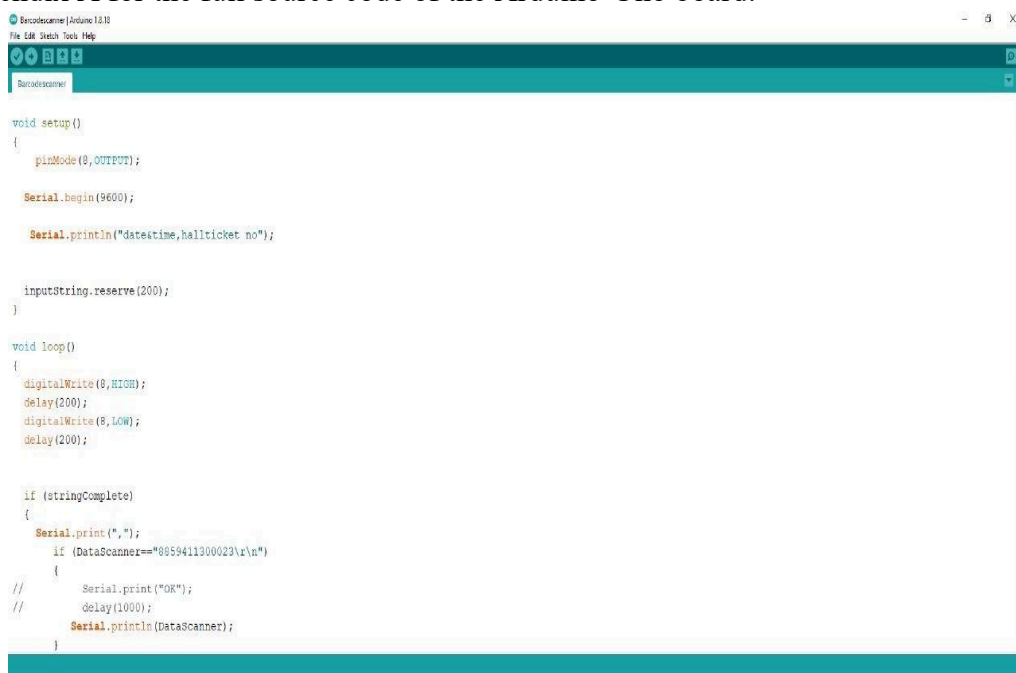
Additional commands are found within the five menus: File, Edit, Sketch, Tools and help.

Programming on Arduino Uno

Fig.2.4 Software IDE

In order for the Arduino-Uno board to be able to interact with the application used in this project certain program (code) needs to be uploaded to the Arduino-Uno.

Arduino Company provides user friendly software which allows writing any code for any function wanted to be performed by the Arduino-Uno and upload it to the board. Refer to appendix A for the full source code of the Arduino-Uno board.

The image shows a screenshot of the Arduino IDE interface. The title bar indicates the file is named 'BarcodeScanner' and the IDE version is 1.8.10. The menu bar includes 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. The toolbar contains icons for opening files, saving, compiling, uploading, and monitoring. The main text area displays the following C++ code:

```
void setup()
{
  pinMode(8, OUTPUT);

  Serial.begin(9600);

  Serial.println("datetime,ballticket no");

  inputString.reserve(200);
}

void loop()
{
  digitalWrite(8,HIGH);
  delay(200);
  digitalWrite(8,LOW);
  delay(200);

  if (stringComplete)
  {
    Serial.print(",");
    if (DataScanner=="8859411300023\r\n")
    {
      Serial.print("OK");
      delay(1000);
      Serial.println(DataScanner);
    }
  }
```

The

Arduino uno is an open-source micro controller board based on microchip ATmega328p micro controller. This is based equipped with set of digital and analog input/output. Early Arduino uno boards used the FTDI USB-to_UART chip and an AT mega168.

CHAPTER 3

CIRCUIT DIAGRAM AND DESCRIPTION

3.1 Working

We have used Arduino and live scanner. Every student is provided with a ID card that consists the unique barcode. They just have to scan their cards using barcode scanner and the system notes down their attendance as per date and time. System then stores all the student attendance records and generate a brief report for admin as required in excel sheet. And this application allow admin to search about a particular student attendance detail in excel sheet.

Such kind of application is very useful in colleges and universities for taking daily attendance as the working of home automation using Arduino is as shown in fig.3.1

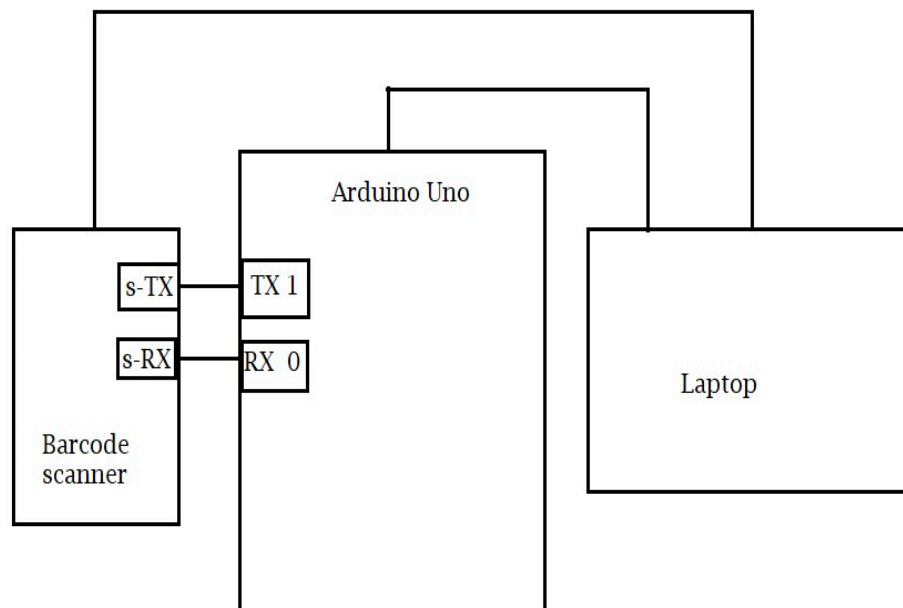
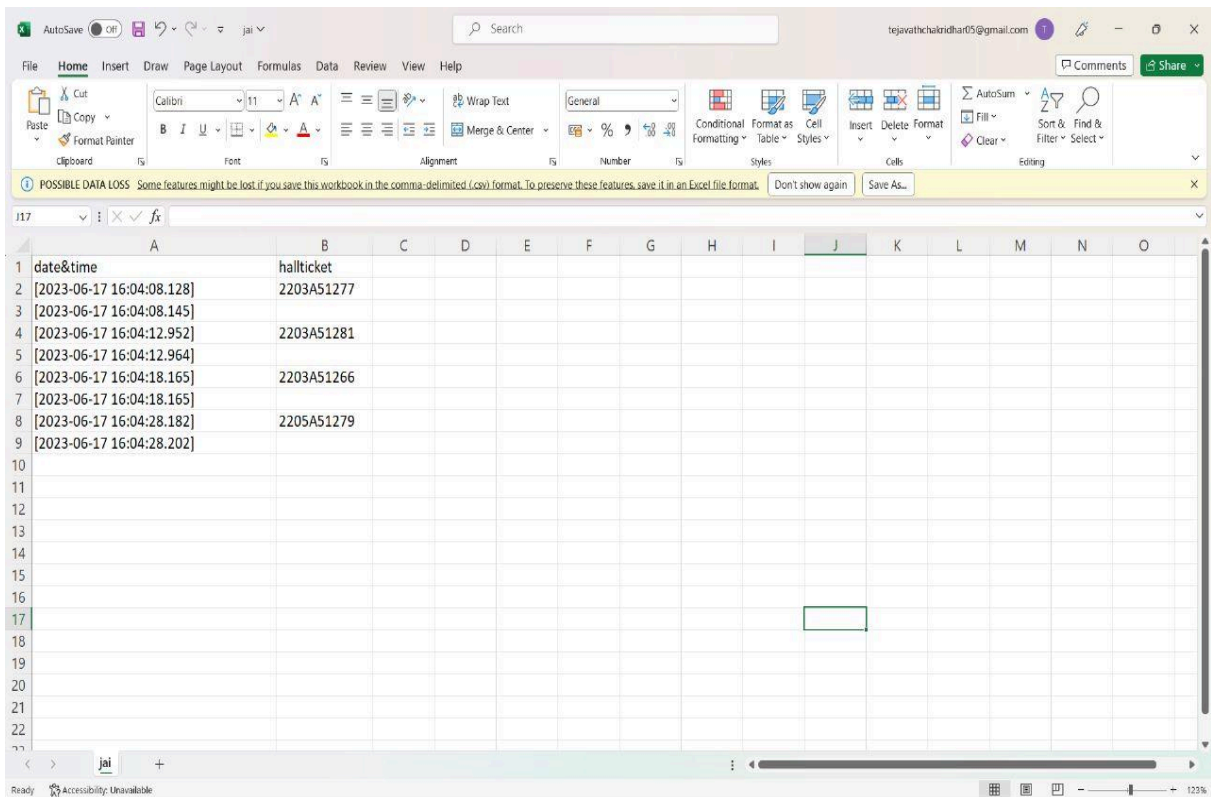


Fig.3.1 Schematic diagram

3.2 RESULTS

The experimental result is as shown in below fig. 3.2



3.2 Experimental result

3.3 ADVANTAGES

- Accuracy
- Time efficiency
- Data analysis
- Integration with other system.
- Real time monitoring.
- Improved accountability
- Attendance notifications.
- Scalability.

3.4 DISADVANTAGES

- Attendance can be marked without the presence of student.
- Equipment and installation cost
- Privacy concerns
- Technical challenges
- Resistance to change
- Initial setup and training
- Maintenance and upgrades
- Integration challenges

CHAPTER 4

CONCLUSION

4.1 CONCLUSION

Based on all the bases I want to conclude that barcode attendance system is most accurate .The developed system presented in this paper has be successfully designed and tested. The students attendance status will be analized and exported. It is an efficient method to store the attendance in the excel sheet rather than wasting the paper .The implemented barcode system offers a convenient and cost-effective method of tracking students attendance and assets in a university setting compared to other technologies. The designed barcode system provides a simple methodology to track student attendance and assets. It has several benefits over the manual methods including the process takes less time and stores the data in electronic mode instead of paperwork. It is considered to be time saving process of taking attendance. If the university administration decides to adopt the barcode system proposed in this paper, the system can be integrated with educate system of the university to automatically capture and update attendance and assets tracking data. The barcode attendance system can be useful for both university and the students.

4.2 FUTURE SCOPE

Now-a-days attendance plays a vital role in academics. In higher education institutions, student participation in the classroom is directly related to their academic performance. However, the majority of student attendance registration is still conventionally done, which is tedious and time-consuming, especially for those courses that involve large numbers of students. Over the years, attendance management has been conducted manually at most of the universities. To overcome the manual attendance issues, we proposed and implemented a smart attendance system with the aim to encourage the potential use of the barcode as a future attendance management system, to track and record student attendance.

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- 12.**

APPENDIX

```
String inputString="";
String DataScanner="";
bool stringComplete=false;

void setup()
{
    pinMode(8,OUTPUT);

    Serial.begin(9600);

    Serial.println("date&time,hallticket no");

    inputString.reserve(200);
}

void loop()
{
    digitalWrite(8,HIGH);
    delay(200);
    digitalWrite(8,LOW);
    delay(200);

    if (stringComplete)
    {
        Serial.print(",");
        if (DataScanner=="8859411300023\r\n")
        {
            //    Serial.print("OK");
            //    delay(1000);
            Serial.println(DataScanner);
        }
    }
}
```

```

// else
// //Serial.print("No Data");
    Serial.println(DataScanner);
//
    inputString="";
    DataScanner="";
    stringComplete=false;
}

}

void serialEvent()
{
    while (Serial.available())
    {

        char inChar=(char)Serial.read();

        inputString+=inChar;

        if (inChar=='\n')
        {
            stringComplete=true;
            DataScanner=String(inputString);

        }
    }
}

```